

Quantitative Aptitude

Simplification

Level-1

Q1 What will come in place of the question mark (?) in the following question.

$$39 \div 13 + 7 - 2 = ?$$

- (A) 8 (B) 6
(C) 4 (D) 2
(E) 1

Q2 What will come in place of the question mark (?) in the following question.

$$\sqrt{144} + 13 = 10 + ?$$

- (A) 10 (B) 15
(C) 18 (D) 20
(E) 25

Q3 $441 \times 2 = ? - 18$

- (A) 700 (B) 900
(C) 800 (D) 600
(E) None of these

Q4 What will come in place of question mark (?) in the following equation?

$$24 \times 45 \div 9 - 60 = ? - 120$$

- (A) 120 (B) 150
(C) 210 (D) 180
(E) 90

Q5 Directions : What should come in place of question mark (?) in the following questions?

$$25 - 13 + 28 = 30 + ?$$

- (A) 10 (B) 5
(C) 15 (D) 0
(E) 2

Q6 Direction : What should come in place of the question mark (?) in the following questions?

$$73 + 73 \div 0.73 = ?$$

- (A) 177 (B) 133
(C) 173 (D) 137
(E) 773

Q7 What will come in the place of question mark (?) in the given expression?

$$(312 \div 26) \times 84 - ? = 314$$

- (A) 690 (B) 694
(C) 698 (D) 682
(E) 622

Q8 What should come in place of the question mark '?' in the following question.

$$\frac{5}{6} \text{ of } 168 - \frac{7}{9} \times 81 = ?$$

- (A) 84 (B) 75
(C) 58 (D) 105
(E) 77

Q9 Directions : What should come in place of question mark (?) in the following questions?

$$321 + 52 \times 5 - 125 = ? \times 12$$

- (A) 64 (B) 46
(C) 54 (D) 38
(E) 30

Q10 Find the value of ?

$$\frac{2}{3} \times \frac{6}{5} \times \frac{10}{4} + \frac{1}{8} - \frac{3}{8} = ?$$

- (A) $\frac{7}{6}$
(B) $\frac{7}{4}$
(C) $\frac{7}{8}$
(D) $\frac{5}{4}$
(E) None of these

Q11 Find the value of given expression

$$121 \div 5 + (8740 \div 5 - 4) \div 5$$

- (A) 373 (B) 343
(C) 370 (D) 473
(E) None of these

Q12 What will come in the place of the question mark

$$8500 + (1600 \div ?) \text{ of } \frac{1}{5} = 8501$$

- (A) None of these (B) 325



- (C) 320 (D) 315
(E) 310

Q13 What should come in place of the question mark '?' in the following question.

$$3685 - 4228 + 3120 \div 2 = ?$$

- (A) 1028 (B) 1017
(C) 1139 (D) 997
(E) **None of these**

Q14 $417 + ?^2 = \frac{2}{5}$ of 1345

- (A) None of these
(B) 9
(C) 60.5
(D) 121
(E) 11

Q15 What will come in the place of question mark (?) in the given expression?

$$\{480 \div 15 + 3 \times (18 \times 12 - 198)\} = ?$$

- (A) 77 (B) 89
(C) 84 (D) 86
(E) 93

Q16 What will come in the place of the question mark '?' in the following question?

$$360 \times 9 \div 3 + 120 + 900 \div 5 = ?$$

- (A) 1426 (B) 1286
(C) 1380 (D) 1275
(E) 1180



Level-2

Q1 What exact value will come in place of (?) in the following questions ?

$$\frac{3}{4} \text{ of } \frac{4}{27} \text{ of } \frac{18}{11} \text{ of } 2673 = ?$$

- (A) 470 (B) 450
(C) 465 (D) 486
(E) 468

Q2 What will come in the place of question mark (?) in the given expression?

$$\left(\frac{5}{9}\right) \times 180 + \left(\frac{3}{7}\right) \times 140 = ? \times 5$$

- (A) 28 (B) 36
(C) 24 (D) 32
(E) 40

Q3 What value will come in the place of the question mark '?' in the following question?

$$2400 \div 12 - \{26 \times 6 + (32 - 5 \times 6)\} = ?$$

- (A) 45 (B) 54
(C) 42 (D) 40
(E) 39

Q4 What will come in the place of question mark (?) in the given expression?

$$12 \times 7 - 36 \div 12 \times 4 = 432 \div 27 + ?$$

- (A) 64 (B) 52
(C) 56 (D) 48
(E) 62

Q5 What will come in place of question mark (?) in the following equation?

$$\{95 - (125 - 64 \div 8 \times 7)\} = ?$$

- (A) 36 (B) 22
(C) 32 (D) 26
(E) 38

Q6 **Directions :** What value will come in place of question mark (?) in the given questions?

$$55\% \text{ of } ? = 45 \times 12 - 36 \times 14 + 8\% \text{ of } 375$$

- (A) 60 (B) 80
(C) 100 (D) 120
(E) None of these

Q7

What will come in the place of question (?) mark in the following number series

$$9^3 + ?^2 \times 25 = 1129$$

- (A) 2 (B) 4
(C) 3 (D) 5
(E) 7

Q8 What will come in place of question mark (?) in the following equation?

$$13^2 - 4^3 - \sqrt{676} + 4 = ?$$

- (A) 73 (B) 79
(C) 83 (D) 89
(E) 93

Q9 **Directions :** What should come in place of question mark (?) in the following questions?

$$(\sqrt[3]{4096} + \sqrt[3]{5832}) \div 17 = (?)^{0.2}$$

- (A) 64 (B) 16
(C) 27 (D) 2
(E) 32

Q10 **Directions:** Read the following question and choose your right answer.

$$5^? \times 5^2 \times 5^5 \div 5^2 = 5^6 \times 5^3$$

- (A) 3 (B) 4
(C) 5 (D) 0
(E) 1

Q11 **Direction: find the simplified value of the following question**

$$150\% \text{ of } 640 + 40 = 120\% \text{ of } ? + 160$$

- (A) 400 (B) 500
(C) 600 (D) 700
(E) 800

Q12 **Direction: find the simplified value of the following question**

$$\frac{4}{5} \text{ of } 1605 + 30\% \text{ of } 5080 = 6589 - ?$$

- (A) 3461 (B) 3781
(C) 3771 (D) 3471
(E) None of these

Q13



Direction: find the simplified value of the following question.

$$\frac{4}{5} \times \left[\frac{1}{2} \text{ of } 25 \right] + 12 \div \frac{3}{4} - 32$$

- (A) -2 (B) -6
(C) 12 (D) 8
(E) 17

Q14 Direction : What exact value will come in the place of question (?) marks:

$$? \% \text{ of } 750 + (15)^2 = (8)^2 + 356$$

- (A) 16 (B) 22
(C) 32 (D) 26
(E) none of these

Q15 What value should come in place of question mark?

$$125\% \text{ of } 444 + \left(\frac{5}{8}\right) \text{ of } 808 - 100 + ? = 1195$$

- (A) 245 (B) 435
(C) 235 (D) 425
(E) None of these

Q16 Which number comes in the place of the question mark

$$140\% \text{ of } 320 - ?\% \text{ of } 400 = 25\% \text{ of } 720$$

- (A) None of these (B) 60
(C) 62 (D) 67
(E) 65

Q17 Which number comes in the place of the question mark.

$$30\% \text{ of } 200 + \sqrt{?} = 48\% \text{ of } 550 - 10\% \text{ of } 150$$

- (A) None of these (B) 35721
(C) 289 (D) 121
(E) 625

Q18 Direction : What exact value will come in the place of question (?) marks:

$$\frac{728}{?} \times 15 + 24\% \text{ of } 550 - 38 = 484$$

- (A) 20 (B) 36
(C) 32 (D) 24
(E) none of these

Q19 Which number comes in the place of question mark

$$(11 \div 2\frac{1}{5}) \div \frac{11}{5} \times 2\frac{1}{2} - 2 = ?$$

- (A) $3\frac{15}{22}$
(B) $\frac{2}{5}$
(C) $\frac{7}{8}$
(D) $7\frac{5}{8}$
(E) $5\frac{1}{3}$

Q20 60% of 55 of 16 - 120% of 410 + 132 ÷ 6 = ? - 10
find the value of ?

- (A) 65 (B) 66
(C) 67 (D) 68
(E) None of these

Q21 Which number comes in the place of the question mark.

$$(\sqrt{8} \times \sqrt{8})^{1/2} + (9)^{1/2} = (?)^3 + \sqrt{8} - 340$$

- (A) None of these (B) 9
(C) 18 (D) 19
(E) 7

Q22 $6 \times \{6 \text{ of } (6 + 6)\} \div \{(6 \div 6) \times (600\% \text{ of } 6)\}$ find the value of the given expression -

- (A) 8 (B) 10
(C) 12 (D) 14
(E) None of these



Level-3

Q1 65% of 400 + $\sqrt{?}$ = 44% of 800 - 12% of 400, find the value of ?.

- (A) 44 (B) 2115
(C) 46 (D) 5
(E) 1936

Q2 Which number comes in the place of the question mark?

$$576 \times 90 \div 24 \div 72 = ? + 44 \div 4 + 10$$

- (A) 7 (B) 8
(C) 9 (D) 10
(E) None of these

Q3 What will come in place of question mark (?) in the given expression?

$$5 \div 10 \text{ of } 10 \times 4 + 4 \div 4 \text{ of } 4 \times 10 + (10 - 4) \div 16 \times 4 = ?$$

- (A) $\frac{41}{5}$
(B) $\frac{31}{5}$
(C) $\frac{22}{5}$
(D) $\frac{21}{5}$

(E) None of these

Q4 What will come in place of question mark (?) in the given expression?

$$22\% \text{ of } 1800 + 55\% \text{ of } 600 - 25\% \text{ of } 1100 - 10\% \text{ of } 1200 = ?$$

- (A) 332 (B) 333
(C) 331 (D) 330
(E) 336

Q5 125% of 90 $\times$ 24% of 900 $\div$ (35 $\times$ 9 $\div$ 7 $\times$ 9 $\times$ 5) = ? simplify the above expression

- (A) 14 (B) 18
(C) 16 (D) 12
(E) 10

Q6 $\left(\frac{1}{4} \times \frac{4}{7} \div \frac{3}{14} \times \frac{7}{3}\right) + \left(\frac{2}{9} \div \frac{4}{3}\right)$ find the value of ?

- (A) $\frac{8}{18}$
(B) $\frac{8}{15}$

(C) $\frac{8}{17}$

(D) $\frac{9}{19}$

(E) None of these

Q7 Directions : what will come in the place of the question mark '?' in the following questions
1000

$$\div 250 = (\sqrt[3]{?} \times \sqrt{1444}) \div (\sqrt[3]{512} \times \sqrt{361})$$

- (A) 3375 (B) 512
(C) 4096 (D) 5412
(E) 2744

Q8 Directions : what will come in the place of the question mark '?' in the following questions

$$14\% \text{ Of } 400 + 16\% \text{ of } 450 + 50\% \text{ of }$$

$$\left\{ \sqrt{9^2} - \sqrt{289} \right\} = ?$$

- (A) 124 (B) 120
(C) 130 (D) 135
(E) None of these

Q9 73 % of 1600 + 32% of 2250 - 12% of 625 = ?³ - 19² - 23 Find the value of ?

- (A) 13 (B) 10
(C) 12 (D) 15
(E) None of these

Q10 Direction: Study the data carefully and answer the following question.

An equation is given below, whose value is 30.

$$(x\% \text{ of } 600 - 8 \times 6) \div 5$$

If value of 'x' were $66\frac{2}{3}\%$ of its original value, then what would be the value of the equation?

- (A) 16.8 (B) 20
(C) 24.2 (D) 18
(E) 12

Q11 Directions : what will come in the place of the question mark '?' in the following questions

$$(27)^2 \times \frac{6}{9} + (7)^3 + 171 = 100 + (?)^3 - 431$$

- (A) 11 (B) 10
(C) 9 (D) 8
(E) 7



Q12 Directions : what will come in the place of the question mark '?' in the following questions

$$(\sqrt{7} - \sqrt{10})^2 + (\sqrt{5} + \sqrt{14})^2 = (?)^3 - 28$$

- (A) 3 (B) 6
(C) 4 (D) 2
(E) None of these



Answer Key

Level-1

Q1 (A)
Q2 (B)
Q3 (B)
Q4 (D)
Q5 (A)
Q6 (C)
Q7 (B)
Q8 (E)

Q9 (D)
Q10 (B)
Q11 (A)
Q12 (C)
Q13 (B)
Q14 (E)
Q15 (D)
Q16 (C)



Level-2

Q1 (D)
Q2 (D)
Q3 (C)
Q4 (C)
Q5 (D)
Q6 (D)
Q7 (B)
Q8 (C)
Q9 (E)
Q10 (B)
Q11 (D)

Q12 (B)
Q13 (B)
Q14 (D)
Q15 (C)
Q16 (D)
Q17 (B)
Q18 (E)
Q19 (A)
Q20 (D)
Q21 (E)
Q22 (C)



Level-3

Q1 (E)
Q2 (C)
Q3 (D)
Q4 (C)
Q5 (D)
Q6 (A)

Q7 (C)
Q8 (A)
Q9 (A)
Q10 (A)
Q11 (A)
Q12 (C)



Hints & Solutions

Level-1

Q1 Text Solution:

$$39 \div 13 + 7 - 2 = ?$$

$$3 + 7 - 2 = ?$$

$$? = 8$$

Q2 Text Solution:

$$\sqrt{144} + 13 = 10 + ?$$

$$12 + 13 = 10 + ?$$

$$25 - 10 = ?$$

$$? = 15$$

Q3 Text Solution:

$$441 \times 2 = ? - 18$$

$$\Rightarrow 882 = ? - 18$$

$$\Rightarrow ? = 882 + 18$$

$$\Rightarrow ? = 900$$

Q4 Text Solution:

Follow BODMAS rule to solve this question, as per the order given below,

Step-1: Parts of an equation enclosed in 'Brackets' must be solved first, and in the bracket,

Step-2: Any mathematical 'Of' or 'Exponent' must be solved next,

Step-3: Next, the parts of the equation that contain 'Division' and 'Multiplication' are calculated,

Step-4: Last but not least, the parts of the equation that contain 'Addition' and 'Subtraction' should be calculated

$$24 \times 45 \div 9 - 60 = ? - 120$$

$$\Rightarrow 24 \times 5 - 60 = ? - 120$$

$$\Rightarrow 120 - 60 = ? - 120$$

$$\Rightarrow 60 = ? - 120$$

$$\Rightarrow 180 = ?$$

Q5 Text Solution:

Given,

$$25 - 13 + 28 = 30 + ?$$

$$40 = 30 + ?$$

$$10 = ?$$

Q6 Text Solution:

$$? = 73 + 100$$

$$? = 173$$

Q7 Text Solution:

$$(312 \div 26) \times 84 - ? = 314$$

$$1008 - 314 = ?$$

$$694 = ?$$

Q8 Text Solution:

$$\frac{5}{6} \text{ of } 168 - \frac{7}{9} \times 81 = ?$$

$$? = 140 - 63$$

$$? = 77$$

Q9 Text Solution:

$$321 + 52 \times 5 - 125 = ? \times 12$$

$$321 + 260 - 125 = ? \times 12$$

$$? \times 12 = 456 = 38$$

Q10 Text Solution:

Applying BODMAS rule

$$\frac{2}{3} \times \frac{6}{5} \times \frac{10}{4} + \frac{1}{8} - \frac{3}{8}$$

$$\frac{4}{5} \times \frac{10}{4} + \frac{1}{8} - \frac{3}{8}$$

$$\frac{5}{5} \times \frac{4}{4} + \frac{1}{8} - \frac{3}{8}$$

$$2 + \frac{1-3}{8}$$

$$2 - \frac{1}{4}$$

$$\frac{7}{4}$$

Q11 Text Solution:

$$121 \div 5 + (1748 - 4) \div 5$$

$$121 \div 5 + 1744 \div 5$$

$$\frac{121}{5} + \frac{1744}{5}$$

$$\frac{1865}{5}$$

$$373$$

Q12 Text Solution:

$$8500 + (1600 \div ?) \text{ of } \frac{1}{5} = 8501$$

$$1600 \div ? \times \frac{1}{5} = 8501 - 8500 = 1$$

$$? = 1600 \div 5 = 320$$

Hence the answer is 320.

Q13 Text Solution:

$$? = 3685 - 4228 + 3120 \div 2$$

$$? = 3685 - 4228 + 1560$$

$$? = 5245 - 4228$$

$$? = 1017$$

Q14 Text Solution:


$$417 + ?^2 = \frac{2}{5} \times 1345$$

$$?^2 = 538 - 417$$

$$? = \sqrt{121} = 11$$

Q15 Text Solution:

$$\{480 \div 15 + 3 \times (18 \times 12 - 198)\} = ?$$

$$32 + 3 \times 18 = ?$$

$$86 = ?$$

Q16 Text Solution:

$$360 \times 9 \div 3 + 120 + 900 \div 5 = ?$$

$$360 \times 3 + 120 + 180 = ?$$

$$1080 + 300 = ?$$

$$? = 1380$$



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Level-2

Q1 Text Solution:

$$? = \frac{3}{4} \times \frac{4}{27} \times \frac{18}{11} \times 2673$$

$$? = 2 \times 243 = 486$$

Q2 Text Solution:

$$\left(\frac{5}{9}\right) \times 180 + \left(\frac{3}{7}\right) \times 140 = ? \times 5$$

$$(5 \times 20) + (3 \times 20) = ? \times 5$$

$$? = 160 \div 5$$

$$? = 32$$

Q3 Text Solution:

$$2400 \div 12 - \{26 \times 6 + (32 - 5 \times 6)\} = ?$$

$$200 - (156 + 2) = ?$$

$$? = 42$$

Q4 Text Solution:

$$12 \times 7 - 36 \div 12 \times 4 = 432 \div 27 + ?$$

$$84 - 12 = 16 + ?$$

$$56 = ?$$

Q5 Text Solution:**Solution (4)**

Follow BODMAS rule to solve this question, as per the order given below,

Step-1: Parts of an equation enclosed in 'Brackets' must be solved first, and in the bracket,

Step-2: Any mathematical 'Of' or 'Exponent' must be solved next,

Step-3: Next, the parts of the equation that contain 'Division' and 'Multiplication' are calculated,

Step-4: Last but not least, the parts of the equation that contain 'Addition' and 'Subtraction' should be calculated

$$\{95 - (125 - 64 \div 8 \times 7)\} = ?$$

$$\Rightarrow \{95 - (125 - 8 \times 7)\} = ?$$

$$\Rightarrow \{95 - (125 - 56)\} = ?$$

$$\Rightarrow \{95 - 69\} = ?$$

$$\therefore ? = 26$$

Q6 Text Solution:

$$\frac{55}{100} \times ? = 45 \times 12 - 36 \times 14 + \frac{8}{100} \times 375$$

$$\Rightarrow \frac{11}{20} \times ? = 540 - 504 + 30$$

$$\Rightarrow ? = \frac{66}{11} \times 20 = 120$$

Q7 Text Solution:

$$9^3 + ?^2 \times 25 = 1129$$

$$729 + ?^2 \times 25 = 1129$$

$$?^2 \times 25 = 400$$

$$?^2 = 16$$

$$? = 4$$

Q8 Text Solution:**Calculation:**

$$13^2 - 4^3 - \sqrt{676} + 4 = ?$$

$$\Rightarrow 169 - 64 - \sqrt{676} + 4 = ?$$

$$\Rightarrow ? = 169 - 64 - 26 + 4$$

$$\Rightarrow ? = 173 - 90 = 83$$

Q9 Text Solution:

$$(16 + 18) \div 17 = (?)^{0.2}$$

$$2 = (?)^{0.2}$$

$$? = 32$$

Q10 Text Solution:

$$5^? \times 5^2 \times 5^5 \div 5^2 = 5^6 \times 5^3$$

$$5^{(?+2+5-2)} = 5^{(6+3)}$$

$$(? + 2 + 5 - 2) = (6 + 3)$$

$$? + 5 = 9$$

$$? = 4$$

Q11 Text Solution:

Let 150% of 640 + 40 = 120% of x + 160

$$960 + 40 = \frac{6x}{5} + 160$$

$$x = \frac{840 \times 5}{6} = 700$$

Q12 Text Solution:

$$\frac{4}{5} \text{ of } 1605 + 30\% \text{ of } 5080 = 6589 - ?$$

$$4 \times 321 + 3 \times 508 = 6589 - ?$$

$$1284 + 1524 = 6589 - ?$$

$$6589 - 2808 = 3781$$

Q13 Text Solution:

$$\frac{4}{5} \times [\frac{1}{2} \times 25] + 12 \times \frac{4}{3} - 32 = \frac{4}{5} \times 2.5 + 16 - 32$$

$$= 10 + 16 - 32$$

$$= -6$$

Q14 Text Solution:

$$\frac{?}{100} \times 750 + (15)^2 = (8)^2 + 356$$



$$\frac{?}{100} \times 750 + 225 = 64 + 356$$

$$\frac{?}{100} \times 750 = 195$$

$$? = \frac{195 \times 100}{750}$$

$$? = 26$$

Q15 Text Solution:

$$125\% \text{ of } 444 + \left(\frac{5}{8}\right) \text{ of } 808 - 100 + ?$$

$$= 1195$$

$$\left(\frac{5}{4}\right) \times 444 + \left(\frac{5}{8}\right) \times 808 - 100 + ?$$

$$= 1195$$

$$555 + 505 - 100 + ? = 1195$$

$$1060 - 100 + ? = 1195$$

$$? = 1195 - 960$$

$$? = 235$$

Q16 Text Solution:

$$140 \times \frac{320}{100} - \frac{?}{100} \times 400 = \frac{25}{100} \times 720$$

$$448 - 4 \times ? = 180$$

$$? = 268 \div 4 = 67$$

Q17 Text Solution:

$$30\% \text{ of } 200 + \sqrt{?} = 48\% \text{ of } 550 - 10\% \text{ of } 150$$

$$200 \times \frac{30}{100} + \sqrt{?} = 550 \times \frac{48}{100} - 150 \times \frac{10}{100}$$

$$60 + \sqrt{?} = \frac{26400}{100} - 15$$

$$\sqrt{?} = 249 - 60$$

$$\sqrt{?} = 189$$

$$? = 35721$$

Q18 Text Solution:

$$\frac{728}{?} \times 15 + \frac{24}{100} \times 550 - 38 = 484$$

$$\frac{728 \times 15}{?} = 484 + 38 - 132$$

$$? = \frac{728 \times 15}{390}$$

$$? = 28$$

Q19 Text Solution:

$$? = \left(11 \times \frac{1}{2\frac{1}{5}}\right) \div \frac{11}{5} \times 2\frac{1}{2} - 2$$

$$? = 5 \div \frac{11}{5} \times \frac{5}{2} - 2$$

$$? = 5 \times \frac{5}{11} \times \frac{5}{2} - 2$$

$$? = \frac{125}{22} - 2 = 3\frac{15}{22}$$

Q20 Text Solution:

$$60\% \text{ of } 55 \text{ of } 16 - 120\% \text{ of } 410 + 132 \div 6 = ? - 10$$

$$\frac{60}{100} \times 55 \times 16 - \frac{120}{100} \times 410 + 132 \div 6 = ? - 10$$

$$3 \times 11 \times 16 - 12 \times 41 + 132 \div 6 = ? - 10$$

$$3 \times 11 \times 16 - 12 \times 41 + 22 = ? - 10$$

$$528 - 492 + 22 = ? - 10$$

$$560 - 492 = ?$$

$$? = 68$$

Q21 Text Solution:

$$(?)^3 + \sqrt{8} - 340 = (\sqrt{8} \times \sqrt{8})^{1/2} + (9)^{1/2}$$

$$(?)^3 = \sqrt{8} + 3 - \sqrt{8} + 340$$

$$(?)^3 = 343$$

$$? = \sqrt[3]{343}$$

$$? = 7$$

Q22 Text Solution:

Given

$$6 \times \{6 \text{ of } (6 + 6)\} \div \{(6 \div 6) \times (600\% \text{ of } 6)\}$$

According to the rule of BODMAS

First solve the brackets in given order $()$, $\{\}$, $[\]$

$$\Rightarrow 6 \times \{6 \text{ of } (12)\} \div \{(6 \div 6) \times (\frac{600}{100} \text{ of } 6)\}$$

$$\Rightarrow 6 \times \{6 \text{ of } 12\} \div \{1 \times 36\}$$

$$\Rightarrow 6 \times 72 \div 36$$

$$\Rightarrow 6 \times 2 = 12.$$

Hence, the value is 12.



Level-3

Q1 Text Solution:

$$400 \times \frac{65}{100} + \sqrt{?} = 800 \times \frac{44}{100} - 400 \times \frac{12}{100}$$

$$\sqrt{?} = 352 - 48 - 260 = 44$$

$$? = (44)^2 = 1936$$

Q2 Text Solution:

Using BODMAS rule

$$576 \times 90 \div 24 \div 72 = ? + 44 \div 4 + 10$$

$$576 \times 0.052 = ? + 11 + 10$$

$$30 = ? + 21$$

$$? = 30 - 21$$

$$? = 9$$

Q3 Text Solution:

$$? = 5 \div 10 \text{ of } 10 \times 4 + 4 \div 4 \text{ of } 4 \times 10 + (10 - 4) \div 16 \times 4$$

$$= 5 \div 100 \times 4 + 4 \div 16 \times 10 + 6 \div 16 \times 4$$

$$= 0.05 \times 4 + 0.25 \times 10 + 0.375 \times 4$$

$$= 0.2 + 2.5 + 1.5$$

$$= 4.2$$

$$= \frac{42}{10}$$

$$= \frac{21}{5}$$

Q4 Text Solution:

$$22\% \text{ of } 1800 + 55\% \text{ of } 600 - 25\% \text{ of } 1100 - 10\% \text{ of } 1200 = ?$$

$$= 22 \times 18 + 55 \times 6 - 25 \times 11 - 10 \times 12 = ?$$

$$= 396 + 330 - 275 - 120 = ?$$

$$= 726 - 395$$

$$= 331$$

Q5 Text Solution:

Applying BODMAS rule

B = Bracket

O = of

D = Division

M = Multiplication

A = Addition

S = subtraction

$$125\% \text{ of } 90 \times 24\% \text{ of } 900 \div (35 \times 9 \div 7 \text{ of } 9 \times 5 = ?$$

$$\frac{125}{100} \times 90 \times \frac{24}{100} \times 900 \div (35 \times 9 \div 7 \times 9 \times 5 = ?$$

$$1.25 \times 90 \times 24 \times 9 = ?$$

$$\div (35 \times \frac{9}{7} \times 9 \times 5)$$

$$1.25 \times 90 \times 24 \times 9 \div (2025) = ?$$

$$1.25 \times 90 \times 24 \times \frac{9}{2025} = ?$$

$$1.25 \times 90 \times 24 \times \frac{1}{225}$$

$$2700 \times \frac{1}{225} = ?$$

$$12 = ?$$

$$? = 12$$

Q6 Text Solution:

$$\left(\frac{1}{4} \times \frac{4}{7} \div \frac{3}{14} \times \frac{7}{3}\right) + \left(\frac{2}{9} \div \frac{4}{3}\right)$$

$$- \left(\frac{7}{9} \times \frac{18}{14}\right) = ? + \frac{5}{18}$$

$$\Rightarrow \left(\frac{1}{4} \times \frac{4}{7} \times \frac{14}{3} \times \frac{7}{3}\right) + \left(\frac{2}{9} \times \frac{3}{4}\right)$$

$$- \left(\frac{7}{9} \times \frac{18}{14}\right) = ? + \frac{5}{18}$$

$$\Rightarrow \frac{14}{9} + \frac{1}{6} - 1 = ? + \frac{5}{18}$$

$$\Rightarrow \frac{28+3-18}{18} = ? + \frac{5}{18}$$

$$\Rightarrow \frac{13}{18} = ? + \frac{5}{18}$$

$$\Rightarrow ? = \frac{13}{18} - \frac{5}{18}$$

$$\Rightarrow ? = \frac{8}{18}$$

Q7 Text Solution:

$$1000/250 = (\sqrt[3]{?} \times \sqrt{1444})$$

$$/ (\sqrt[3]{512} \times \sqrt{361})$$

$$4 \times 8 \times 19 = \sqrt[3]{?} \times 38$$

$$? = 16^3$$

$$? = 4096$$

Q8 Text Solution:

$$400 \times \frac{14}{100} + 450 \times \frac{16}{100} + \frac{1}{2} \left\{ \sqrt{9^2} - \sqrt{289} \right\}$$

$$=?$$

$$56 + 72 - \frac{8}{2} = ?$$

$$56 + 72 - 4 = ?$$

$$? = 124$$

Q9 Text Solution:

$$73\% \text{ of } 1600 + 32\% \text{ of } 2250 - 12\% \text{ of } 625 = ?^3 - 19^2 - 23$$

$$\frac{73}{100} \times 1600 + \frac{32}{100} \times 2250 - \frac{12}{100} \times 625$$

$$= ?^3 - 361 - 23$$

$$-361-23$$

$$73 \times 16 + 16 \times 45 - 3 \times 25 = ?^3 - 384$$

$$1168 + 720 - 75 = ?^3 - 384$$



$$1888 - 75 = ?^3 - 384$$

$$1813 = ?^3 - 384$$

$$?^3 = 1813 + 384$$

$$?^3 = 2197$$

$$? = 13$$

Q10 Text Solution:

Since,

$$(x\% \text{ of } 600 - 8 \times 6) \div 5 = 30$$

$$(6x - 48) = 150$$

$$x = 33$$

If value of 'x' were $66\frac{2}{3}\%$ of its original value.

So, the new value of 'x' would be = $66\frac{2}{3}\%$ of 33

$$= 22$$

And the value of the equation would be:

$$(22\% \text{ of } 600 - 8 \times 6) \div 5 = 16.8$$

Q11 Text Solution:

$$27 \times 27 \times 6/9 + 343 + 171 - 100 + 431$$

$$= (?)^3$$

$$486 + 945 - 100 = (?)^3$$

$$1331 = (?)^3$$

$$? = 11$$

Q12 Text Solution:

$$(?)^3 = 7 + 10 - 2\sqrt{70} + 5 + 14 + 2\sqrt{70}$$

$$+ 28$$

$$(?)^3 = 36 + 28$$

$$\therefore ? = \sqrt[3]{64} = 4$$



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